NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR STAR - A & APEX, ALL IITS | DT: 08/01/2026
Time: 3 Hrs | GTM – 12 | Max. Marks: 300

KEY SHEET

PHYSICS

01	02	03	04	05	06	07	08	09	10
3	4	4	2	1	1	1	2	3	1
11	12	13	14	15	16	17	18	19	20
3	3	2	2	2	2	1	4	1	1
21	22	23	24	25					
3	3	11	8	15					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
2	3	4	3	2	4	1	3	4	3
36	37	38	39	40	41	42	43	44	45
1	3	4	4	2	1	3	4	4	3
46	47	48	49	50					
4	0	5	1000	3647					

MATHEMATICS

51	52	53	54	55	56	57	58	59	60
2	4	2	2	4	1	3	3	4	1
61	62	63	64	65	66	67	68	69	70
1	4	2	4	2	3	4	1	2	4
71	72	73	74	75					
405	1421	8	16	75					

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HINTS
PHYSICS

1. Conceptual
2. Conceptual
3. The fringe – width β is given by

$$\beta = \frac{\lambda D}{2d} \text{ and } \beta' = \frac{\lambda D'}{2d}$$

Where λ is the wavelength of light used, D is the distance of the screen from the two slits and 2 d is the separation between two slits.

Now $\beta - \beta' = \frac{\lambda(D - D')}{2d}$

$$\Rightarrow \lambda = \frac{(\beta - \beta')2d}{D - D'}$$

$$\lambda = \frac{\left(3\times10^{-5}\right)\left(10^{-3}\right)}{5\times10^{-2}}$$

$$= 0.6\times10^{-6} m = 6000\,\mathring{A}$$

4. In the steady state current in the branch having capacitor is zero. The effective circuit is shown in the figure. Current through the cell

$$I = \frac{3}{7}A$$

Therefore current through the voltmeter is $I' = \frac{2}{3}I = \frac{2}{7}A$

Therefore, voltmeter reading is $= I'R_V = \frac{400}{7}V$

5. $$2.45\times10^{-2} = \frac{2}{9}\frac{\left(1.8\times10^{-3}\right)^2\times6900}{\eta}\times9.8$$

$$\eta = 1.99$$

6. Initially, $P = \frac{V_{rms}^2}{R} \Rightarrow V_{rms} = \sqrt{PR}$

Finally, $P' = I_{rms}^2 R = \frac{V_{rms}^2}{z^2}R = \frac{PR}{z^2}\times R = P\left(\frac{R}{z}\right)$

7. $$\frac{1}{2}\times10\times3^2 = \frac{1}{2}\times15\times2^2 + \frac{1}{2}\times3000\times x^2$$

$$30 = 3000\times x^2$$

$$x = 0.1$$

8. From snell's law

$$\mu\sin\theta = 16\mu\sin x$$

$$\Rightarrow \sin x = \frac{5}{8}\sin\theta$$

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9. Conceptual

10. Conceptual

11. In one half – life the number of active nuclei reduces to half the original number. Thus, in two half – lives the number is reduced to $\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)$ of the original number. The number of remaining active nuclei is , therefore,

$$6.0\times10^{18}\times\left(\frac{1}{2}\right)\times\left(\frac{1}{2}\right)=1.5\times10^{18}$$

12. At any time

$$F_{thrust}-F_{friction}=ma$$

$$\rho av^2-\mu mg=ma$$

$$\rho a\left(2gh\right)-\left(\frac{a}{A}\right)\left(\rho Ab\right)g=\left(\rho Ah\right)a'$$

$$2\rho agh-\rho agh=\rho Aha';\frac{a}{A}g=a'$$

13. $$V=\frac{kQ}{\frac{R}{2}}+\frac{kQ}{R}=\frac{Q}{2\pi\varepsilon_0 R}+\frac{Q}{4\pi\varepsilon_0 R}$$

14. $$\because\ f=n\frac{v}{2\ell}$$

$$\Rightarrow 1100=n\times\frac{330}{2\times30\times10^{-2}}$$

This shows that air column vibrates in 2nd harmonic. This implies that wavelength

$$\lambda=\ell=30cm$$

$$\because\ \Delta P=0.1P_0\sin kx\cos wt$$

x is measure from open end.

At x=5cm, i.e ., at $x=\frac{\lambda}{6}$

$$\Delta P=0.1P_0\sin\left(\frac{2\pi}{\lambda}.\frac{\lambda}{6}\right)\cos wt$$

$$\Rightarrow P_{\max}-P_{\min}=2\times0.1P_0.\frac{\sqrt{3}}{2}$$

$$=\sqrt{3}\times10^4\,N/m^2$$

15. Conceptual

17. Voltage across capacitor will lag current by 90^0;

$$i_C=\frac{200\sqrt{2}}{10}\sin\left(\omega t+\frac{\pi}{4}+\frac{\pi}{2}\right)$$

$$i=20\sqrt{2}\sin\left(\omega t+\frac{3\pi}{4}\right)$$

18. Conceptual

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19. KE of the vessel $\frac{1}{2}Mv^2$

When the vessel is suddenly stopped, the ordered motion of the gas molecules is converted into disordered motion of the molecules, thereby increasing the internal energy of the gas. Thus,

$$\Delta U = nC_V\Delta T = \frac{1}{2}mv^2 = \frac{1}{2}(nM)v^2$$

Where, n is number of moles of the gas in the vessel and M is molecular weight of the gas.

$$\therefore \Delta T = \frac{Mv^2}{2C_V}$$

As, $C_V = \frac{R}{\gamma - 1}$

$$\Delta T = \frac{Mv^2(\gamma - 1)}{2R}$$

20. Conceptual

21. Energy required removing the electron from the third orbit

$$= \frac{13.6}{3^2}eV = 1.5eV = 1.5\times 1.6\times 10^{-19}\ J = 3eV$$

23. $$R_{cm} = \frac{u^2\sin 20}{g} = 5\sqrt{3}$$

Let M be the mass of each fragment

$$\Rightarrow \frac{M\sqrt{3} + (n-1)M\times 7\sqrt{3}}{nM} = 5\sqrt{3}$$

$$\Rightarrow n = 3$$

24. Conceptual

25. Conceptual

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CHEMISTRY

26. The E_a of forward reaction is more then that of backward reaction

K_9 decreases with temperature. Hence ΔH must be –ve as evident from the Eq.

$$\frac{d \ln K_9}{dT} = \frac{\Delta H}{RT^2}$$

$$\Delta H = E_a(FR) - E_a(BR)$$

$$\therefore E_a(FR) < E_a(BR)$$

29. MnO_4^- is stronger oxidizing agents than $Cr_2O_7^{-2}$. MnO_4^- has more SRP

30. $2Al + 6HCl \rightarrow 2AlCl_3 + 3H_2$

$27 \times 2 \rightarrow 3 \times 22.4$

$x \rightarrow 22.4$

$$x = \frac{22.4 \times 27 \times 2}{3 \times 22.4} = 18$$

35. key 3

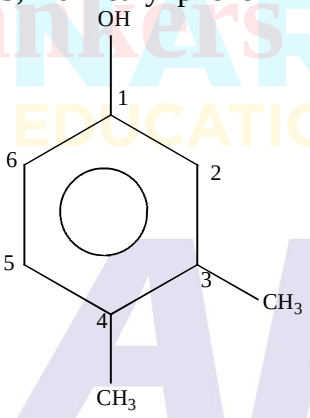
Group -11 - Rg

Group -12 - Cn

Group -13 - Nh

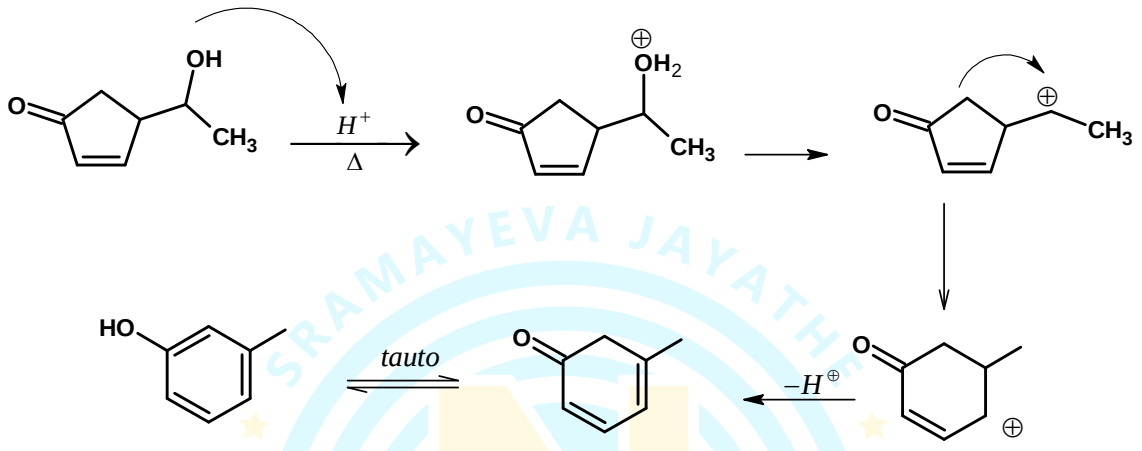
Group -14 – Fl

36. 3, 4 dimethyl phenol



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37.



38. $\frac{p_1v_1}{T_1} = \frac{p_2v_2}{T_2}$

$\frac{700 \times 120}{300} = \frac{760 \times v_2}{273}$

$P_{dry\ gas} = P_{moist\ gas} - Aques\ tension$

$P_{dry\ gas} = 715 - 15 = 7000$

$v_2 = 100.578$

$\%N = \frac{28}{gm} \times V_{N_2}\ initial\ at\ S.T.r \times \frac{22400}{100}$ *wt toO.C in*

$\frac{28}{22400} \times \frac{100.578}{0.8} \times 100 = 15.71\%$

39. $\sqrt{\ell(\ell+1)}\frac{h}{2\pi}$

For d $\ell = 2 \Rightarrow \frac{h}{2\pi}\sqrt{2(2+1)} = \frac{h}{2\pi}\sqrt{6}$

For f $\ell = 3 \Rightarrow \frac{h}{2\pi}\sqrt{3(3+1)} = \frac{h}{2\pi}\sqrt{12}$

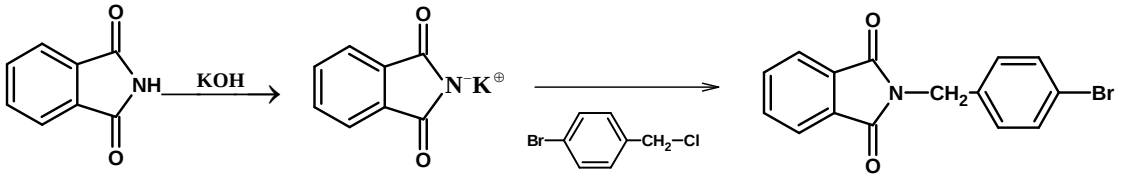
40. CO_2

NO, O_2^-, O_2 are paramagnetic according to

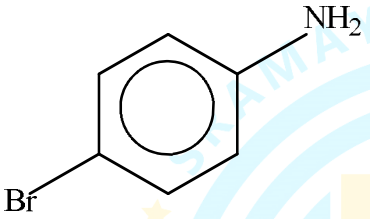
MOT, unparired $\bar{e}$ are in $\sigma^*_{px}, \sigma^*_{py}$

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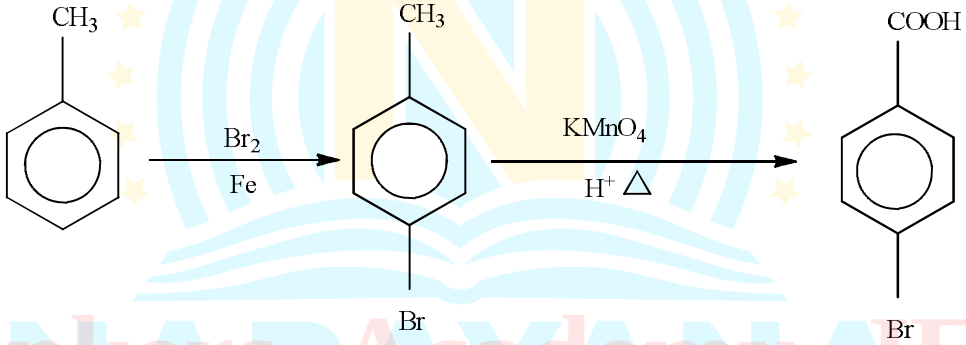
41.



.



42.



43. $NaOH + HCl \rightarrow NaCl + H_2O$

$100\times0.3 \quad 50\times0.6 \quad 30$ m mol

30 30

Concentration of $NaCl = \frac{3}{100} = .003$

45. at equivalence point

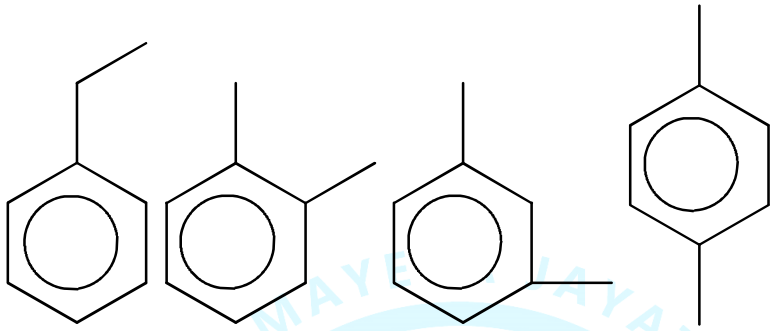
m. eq. of $NaOH$ = m. eq. of H_2SO_4

$40\times M_{NaOH} = 2\times0.01\times20$

$M_{NaOH} = \frac{2\times0.01\times20}{40} = 0.01$

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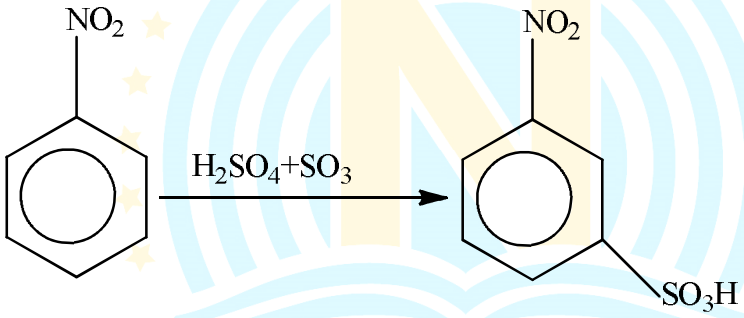
46.



47. $Zn^{+2} \rightarrow 3d^{10}4S^{0}$

Zinc has low enthalpy of atomization

48.



$123\,gms \rightarrow 203\,gms$

$x \rightarrow 8.25\,gms$

$x = \dfrac{123\times 8.25}{203} = 4.99 \equiv 5$

49. $n = \dfrac{1}{100} = 0.01$

$\Delta H_C = \dfrac{c\times\Delta T}{n} = \dfrac{2.5\times 4}{0.01} = 1000\,\text{kJ}$

50. $A \rightarrow B = \ -P\Delta V = -15(6-2) = -60\,lit\,atm$

$B \rightarrow C$ Since volume in constant nowork is done

$C \rightarrow D \ -P\Delta V = -6(2-6) = +24\,lit.atm$

$D \rightarrow A$ Volume is constant w=0

Total work done $= -60 + 24$

$= -36\times 101.3$

-3646.8 J

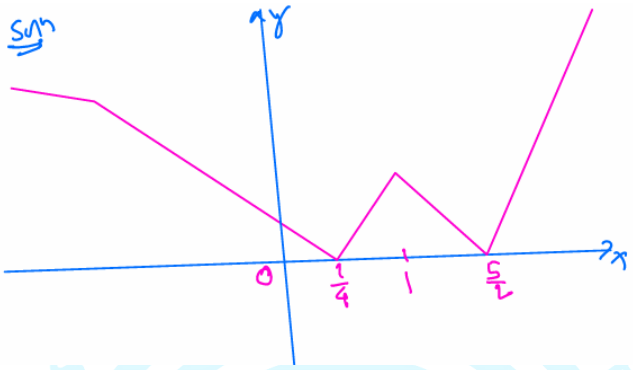
$-(-3646.8)$

$= 3646.8 \equiv 3647$

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MATHEMATICS

51.



Critical points are $-2, 1, \frac{5}{2}$

Points of local maxima $x = 1$

Points of local maxima $x = \frac{1}{4}, \frac{5}{2}$

So $k = 1$ *and* $L = 2$

52. Let direction cosines of a line are $\cos\alpha, \cos\beta, \cos\gamma$

So $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

But $\beta = \gamma = \frac{\alpha}{2}$

So $\cos^2\alpha + 2\cos^2\frac{\alpha}{2} - 1 = 0$

$\cos^2\alpha + \cos\alpha = 0$

$\Rightarrow \cos\alpha = 0, -1$

If $\Rightarrow \cos\alpha = 0 \Rightarrow \alpha = \frac{\pi}{2}$

$\Rightarrow \beta = \gamma = \frac{\pi}{4}$

If $\cos\alpha = -1 \Rightarrow \alpha = \pi$

$\Rightarrow \beta = \gamma = \frac{\pi}{2} \quad \therefore |\beta_1 - \beta_2| + |\gamma_1 - \gamma_2| = \frac{\pi}{4} + \frac{\pi}{4} = \frac{\pi}{2}$

53. Family of circles passing through $A(0,1)$ and $B(3,0)$ is

$x(x-3) + y(y-1) + \lambda(x + 3y - 3) = 0$

As it passes through $c(1,4)$

$\therefore 1(-2) + 4(3) + \lambda(1 + 12 - 3) = 0$

$10 + \lambda(10) = 0$

$\lambda = -1$

So circle is

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$x(x-3)+y(y-1)-(x+3y-3)=0$

$D(k,k+1)$ will satisfy it

$k(k-3)+(k+1)k-(k+3(k+1)-3)=0$

$2k^2-6k=0$

$k=0,3$ but $k=0$ does not give distinct point $\therefore k=3$

54. mean $\overline{x}=\dfrac{1+2+a+4+b}{5}=4$

$a+b=13$ ....(i)

Variance $\sigma^2=\dfrac{\sum x_i^2}{n}-(\overline{x})^2=6$

$=\dfrac{1^2+2^2+a^2+4^2+b^2}{5}-16=6$

$a^2+b^2=89$ .....(ii)

$(a+b)^2=a^2+b^2+2ab$

$169=89+2ab$

$80=2ab$

ab=40 ……(iii)

From (i) and (ii) a=8 and b=5

The observations X_i are

1, 2, 8, 4, 5

New observations are

$x_1+1,\ x_2+2,\ x_3+3, x_4+4, x_5+5$

$2,4,11,8,10$

Mean of new observations $=\dfrac{2+4+11+8+10}{5}$

$=7$

Variance of new observations

$=\dfrac{2^2+4^2+11^2+8^2+10^2}{5}-(7)^2$

$=\dfrac{4+16+121+64+100}{5}-49$

$=61-49$

$=12$

55. Let the family of lines be

$\sin\theta(2x+3y-5)+\cos\theta(3x-2y-1)=0$

These lines passes through intersection of the lines

$2x+3y-5=0$

And $3x-2y-1=0$

$\Rightarrow$ point of intersection is $P(1,1)$

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$d = PQ = \sqrt{(2-1)^2 + (5-1)^2} = \sqrt{1+16}$

$\therefore d^2 = 17$

56. $A = \{-4,-3,-2,-1,0,1,2,3\}$

$R = \{(-4,-4),(-3,-3),(-2,-2),(-1,-1),(0,0),(1,1),(2,1),(3,1)\}$

$P = 8$

$q = 2 \quad \{Add\ (2,\ 2),\ (3,\ 3)\}$

$r = 2 \quad \{Add\ (1,\ 2),\ (1,\ 3)\}$

$2p + q + 3r = 16 + 2 + 6 = 24$

57. $D = 0$

$16(k-1)^2 = 4(3k-2)$

$4k^2 - 8k + 4 = 3k - 2$

$4k^2 - 11k + 6 = 0$

$4k^2 - 8k - 3k + 6 = 0$

$4k(k-2) - 3(k-2) = 0$

$(k-2)(4k-3) = 0$

So $k = 2, \frac{3}{4}$

$k = 2$, since roots are positive

So $p(2,3)$

$d = \left|\frac{4(2) - 3(3) + 16}{5}\right| = 3$

58. Number of triangle are ${}^{11}C_1 . {}^{10}C_2 + {}^{11}C_2 . {}^{10}C_1 + {}^{1}C_1 . {}^{11}C_1 . {}^{10}C_1 = 495 + 550 + 110 = 1155$

59. Let $I = \int_0^{\pi} \frac{2x\,dx}{8\sin^2 x + \cos^2 x} \quad .....(i)$

Using property $\int_0^a f(x)dx = \int_0^a f(a-x)dx$

(i) + (ii) $\Rightarrow$

$2I = \int_0^{\pi} \frac{2\pi\,dx}{8\sin^2 x + \cos^2 x}$

Apply property $\int_0^{2a} f(x)dx = 2\int_0^a f(2a-x)dx \qquad (\because f(x) = f(2a-x))$

$I = 2\pi \int_0^{\pi/2} \frac{dx}{8\sin^2 x + \cos^2 x}$

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$$= 2\pi \int_0^{\pi/2} \frac{\sec^2 x dx}{8\tan^2 x + 1}$$

Put $\tan x = t$

$\sec^2 x dx = dt$

$$= \frac{2\pi}{8} \int_0^{\infty} \frac{dt}{t^2 + \left(\frac{1}{2\sqrt{2}}\right)^2}$$

$$= \frac{2\pi}{8}.2\sqrt{2}\left(\tan^{-1} 2\sqrt{2}t\right)_0^{\infty}$$

$$= \frac{4\sqrt{2}\pi}{8} \qquad \frac{\pi}{2}$$

$$= \frac{\sqrt{2}\pi^2}{4} = \frac{\pi^2}{2\sqrt{2}}$$

60. Put

$x = 2 \quad f(2) + 5f(18) = 6 \quad .....(i)$

$x = 18 \quad f(18) + 5f(2) = 54 \quad .....(ii)$

(i) + (ii) give $6\left(f(2) + f(18)\right) = 60$

So $f(2) + f(18) = 10$

Put x=6

$f(6) + 5f(6) = 18$

$6f(6) = 18$

$f(6) = 3$

61. Area $= \int_0^4 \left(2\sqrt{x} - x\right) dx$

$$= \left(\frac{2x^{3/2}}{3/2} - \frac{x^2}{2}\right)_0^4$$

$$= \left(\frac{4}{3}\right)8 - 8$$

$$= \frac{8}{3} \text{ sq units}$$

62. $1 + \log_{0.5}\left(x^2 - 5x + 6\right) > 0$

$\Rightarrow 1 - \log_2\left(x^2 - 5x + 6\right) > 0$

$\Rightarrow \log_2\left(x^2 - 5x + 6\right) < 1$

$\Rightarrow 0 < x^2 - 5x + 6 < 2$

$\Rightarrow x^2 - 5x + 6 > 0$ *and* $x^2 - 5x + 4 < 0$

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$\Rightarrow x \in (1,2) \cup (3,4)$

$\therefore \alpha + \beta + \gamma + \delta = 10$

63. $\sum_{x=0}^{\infty} P(x) = 1$

$4 + \frac{7}{2} + \frac{10}{2^2} + \frac{13}{2^3} + \ldots\ldots = \frac{1}{\lambda}$

$\frac{4}{2} + \frac{7}{2^2} + \frac{13}{2^3} + \ldots\ldots = \frac{1}{2\lambda}$

$4 + \frac{3}{2} + \frac{3}{2^2} + \frac{3}{2^3} + \ldots\ldots = \frac{1}{\lambda} - \frac{1}{2\lambda}$

$4 + \frac{3}{2}(2) = \frac{1}{2\lambda}$

$\Rightarrow \lambda = \frac{1}{14}$

$\therefore P(x \geq 2) = 1 - P(0) - P(1)$

$= 1 - \frac{1}{14}\left(4 + \frac{7}{2}\right)$

$= 1 - \frac{15}{28} = \frac{13}{28}$

64. $\frac{dy}{dx} - 2(\cos ec^2 x)\, y = \cos ec^2 x$

IF= $e^{-\int 2\cos ec^2 x dx} = e^{2\cot x}$

$\therefore$ solution is

$y.e^{2\cot x} = \int e^{2\cot x}.\cos ec^2 x\, dx$

Put $\cot x = t$

$-\cos ec^2 x dx = dt$

$y.e^{2\cot x} = -\int e^{2t} dt$

$= -\frac{e^{2t}}{2} + c$

$y = -\frac{1}{2} + ce^{-2\cot x}$

$y\left(\frac{\pi}{2}\right) = -\frac{1}{2} + c$

$\therefore c = e^2$

So solution is $y = \frac{-1}{2} + e^2.e^{-2\cot x}$

$\therefore y\left(\frac{\pi}{4}\right) = -\frac{1}{2} + e^2.e^{-2} = \frac{1}{2}$

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65. $Z_1 + Z_2 + Z_3 = 3Z_0$

And $\sum z_1^2 = \sum z_1 z_2$

$z_1^2 + z_2^2 + z_3^2 + 2\left(z_1^2 + z_2^2 + z_3^2\right) = 9z_0^2$

$z_1^2 + z_2^2 + z_3^2 = 3z_0^2$

So $\sum_{k=1}^{3}\left(2z_k - z_0\right)^2 = 4\left(z_1^2 + z_2^2 + z_3^2\right)$

$-4z_0\left(z_1 + z_2 + z_3\right) + 3z_0^2$

$\therefore \sum_{k=1}^{3}\left(2z_k - z_0\right)^2 = 4\left(3z_0^2\right) - 4z_0\left(3z_0\right) + 3z_0^2$

$= 3z_0^2$

$= 3\left(1+i\right)^2$

$= 6i$

66. $2\left(1+\sqrt{3}\right)\sin x + \sqrt{3}\left(1-\sin^2 x\right) = 4 + \sqrt{3}$

$\sqrt{3}\sin^2 x - 2\left(1+\sqrt{3}\right)\sin x + 4 = 0$

$\sqrt{3}\sin^2 x - 2\sqrt{3}\sin x - 2\sin x + 4 = 0$

$\sqrt{3}\sin x\left(\sin x - 2\right) - 2\left(\sin x - 2\right) = 0$

$\left(\sqrt{3}\sin x - 2\right)\left(sinx - 2\right) = 0$

$\Rightarrow \sin x = \dfrac{2}{\sqrt{3}}$

$\therefore$ Number of solutions are 7

67. $ae = 4$

$a = 8$

$b = a\sqrt{1-e^2}$

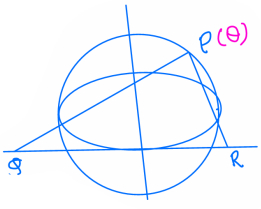
$= 8\sqrt{1-\dfrac{1}{4}}$

$= 4\sqrt{3}$

Area of ΔPQR

$= \dfrac{1}{2}(4b)\left(a\sin\theta + b\right)$

Maximum Area $= 2b\left(a+b\right) = 32\sqrt{3}\left(2+\sqrt{3}\right)$



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68. Let $P\left(x_1, \frac{x_1^2}{20}\right)$

Shortest distance $= CP - CQ$

$C(15,0)$ center of circle

$CQ =$ radius of circle = 1

Let $\ell = CP^2$

$= (x_1 - 15)^2 + \frac{x_1^4}{400}$

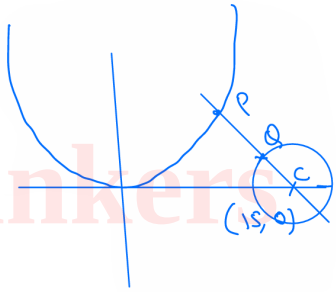
$\frac{d\ell}{dx_1} = 2(x_1 - 15) + \frac{4x_1^3}{400}$

So $\frac{d\ell}{dx_1} = 0 \Rightarrow x_1 = 10$

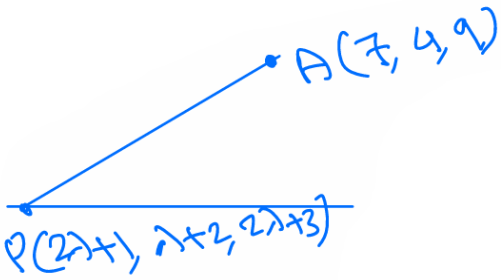
$\therefore P(10,5)$

$CP = \sqrt{25+25} = 5\sqrt{2}$

Shortest distance $PQ = 5\sqrt{2} - 1$



69.



Equation of given line is $\frac{x-1}{2} = \frac{y-2}{1} = \frac{z-3}{2}$

So a point P on it is

$P(2\lambda+1, \lambda+2, 2\lambda+3)$

Also AP is parallel to

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$$\frac{x+5}{3}=\frac{y+1}{2}=\frac{z-\frac{7}{2}}{3}$$

So $\frac{2\lambda-6}{3}=\frac{\lambda-2}{2}=\frac{2\lambda-6}{3}$

$4\lambda-12=3\lambda-6$

$\lambda=6$

So $P(13,8,15)$

$\therefore AP=\sqrt{36+16+36}=\sqrt{88}$

$\therefore d^2=88$

70. $Tr=\frac{1+2+2^2+.....+2^{r-1}}{r!}$

$=\frac{2^r-1}{r!}$

$S_\infty=\sum_{r=1}^{\infty}\frac{2^r}{r!}-\sum_{r=1}^{\infty}\frac{1}{r!}$

$=\left(e^2-1\right)-(e-1)$

$=e^2-e$

71. $|A|=\begin{vmatrix}3 & 4 & P\\ 2 & 5 & -1\\ 3 & 6 & 2\end{vmatrix}$

$=3(16)-4(7)+P(-3)=-1$

$P=7$

$|A|=-1\Rightarrow A^{-1}=-AdjA$

$\left(A^{-1}\right)^{2026}=AdjA^{2026}$

So $B^{-1}=Adj\left(A^{2025}\left(A^{-1}\right)^{2026}\right)$

$=Adj\left(A^{-1}\right)=Adj\left(-AdjA\right)$

$=|A|A$

$=-A$

$\therefore B=(-A)^{-1}=-A^{-1}=AdjA$

Let $C=PB+2I$

$=7B+2I$

$=7Adj\ A+2I$

$\therefore AC=7|A|I+2A$

$2A-7I$

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$$= \begin{pmatrix} 6 & 8 & 14 \\ 4 & 10 & -2 \\ 6 & 12 & 4 \end{pmatrix} - \begin{pmatrix} 7 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 7 \end{pmatrix}$$

$$= \begin{pmatrix} -1 & 8 & 14 \\ 4 & 3 & -2 \\ 6 & 12 & -3 \end{pmatrix}$$

$|A||C| = -1(15) - 8(0) + 14(30)$

$\therefore |C| = -405$

72. Put $x = 1$ we get

$a_0 + a_1 + a_2 + ...a_{24} = 3^{12}$

put $x = -1$ we get

$a_0 - a_1 + a_2 - a_3 + ... = 1$

$\therefore a_1 + a_3 + a_5 + ...a_{23} = \dfrac{3^{12} - 1}{2}$

$= \dfrac{\left(3^6 - 1\right)\left(3^6 + 1\right)}{2}$

$= 728 \times 365$

Also $\left(1 + \left(1 + x\right)\right)^{12} = 1 + {}^{12}C_1 x\left(1 + x\right) + {}^{12}C_2 x^2\left(1 + x\right) + ...$

$\therefore a_2 = 12 + {}^{12}C_2 = 78$

So $\dfrac{a_1 + a_3 + a_5 + ...a_{23} - 91a_2}{182} = k$

$\dfrac{728 \times 365 - 91 \times 78}{182}$

$365 \times 4 - 39 = 1421$

73. Centre of hyperbola in $(5, 4)$

Equation of hyperbola is $\dfrac{(x-5)^2}{a^2} - \dfrac{(y-4)^2}{9-a^2} = 1$

$\left(9 - a^2\right)(x-5)^2 - a^2(y-4)^2 = a^2\left(9 - a^2\right)$

$\left(9 - a^2\right)\left(x^2 - 10x + 25\right) - a^2\left(y^2 - 8y + 16\right) = a^2\left(9 - a^2\right)$

By comparing coeff of x^2

$a^2 = 1$, $\alpha = 80$, $\beta = 8$, $\gamma = 176$

So $2\alpha + 3\beta - \gamma = 160 + 24 - 176$

$= 8$

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74. $\lim\limits_{x\to 0}\left(\dfrac{\tan^{-1}x}{x}\right)^{\frac{1}{\sin^2 x}} = e^{\lim\limits_{x\to 0}\left(\frac{\tan^{-1}x}{x}-1\right)\frac{1}{\sin^2 x}}$

$= e^{\lim\limits_{x\to 0}\dfrac{\tan^{-1}x - x}{x.\frac{\sin^2 x}{x^2}.x^2}}$

$= e^{\lim\limits_{x\to 0}\dfrac{\tan^{-1}x - x}{x^3}}$

$= e^{\lim\limits_{x\to 0}\dfrac{x-\frac{x^3}{3}x}{x^3}} = e^{-\frac{1}{3}}$

So $k = e^{-\frac{1}{3}} \therefore 48\log e^k = -\dfrac{48}{3} = -16$

75. Given $\vec{b}\times\vec{d} = \vec{c}\times\vec{d}$

Or $\left(\vec{b}-\vec{c}\right)\times\vec{d} = 0$

$\Rightarrow \vec{d}$ is parallel to $\vec{b}-\vec{c}$

So $\vec{d} = \lambda\left(\vec{b}-\vec{c}\right)$

Taking dot product with $\vec{a}$

$\vec{a}.\vec{d} = \lambda\left(\vec{a}.\vec{b} - \vec{a}.\vec{c}\right)$

$9 = \lambda(-2-7) \Rightarrow \lambda = -1$

So $\vec{d} = \vec{c} - \vec{b} = \hat{i} + 3\hat{j} - 4\hat{k}$

Now use $\left(\vec{a}\cdot\vec{d}\right)^2 + \left(\vec{a}\times\vec{d}\right)^2 = |\vec{a}|^2\,|\vec{d}|^2$

$81 + \left(\vec{a}\times\vec{d}\right)^2 = 6.(26)$

$= 156$

$\therefore \left(\vec{a}\times\vec{d}\right)^2 = 156 - 81 = 75$

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